

Anant Khanna

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EDUCATION

University of British Columbia

Bachelor of Applied Science in **Computer Engineering** - Dean's List 2025

Vancouver, BC

Sept. 2024 – May 2028

EXPERIENCE

Software Engineer | Agriprix

May 2026 – Present

- Rebuilt the **hybrid intent router** (deterministic + LLM classification) of an LLM agent platform spanning **70+ tools**; added **per-call token/latency observability** with **multi-provider failover**
- Implemented **Stripe authorize/capture** checkout with **deterministic idempotency keys**, an **integer-cent commission engine**, and **per-farm fulfillment state machines** for multi-supplier orders
- Led a security audit surfacing **67 vulnerabilities (5 critical)**: patched **mass-assignment/IDOR** serializers, fixed **overselling races** via row-level locking, and shipped a **Playwright E2E suite**

Machine Learning Software Engineer Team Lead | UBC UAS

Sept. 2024 – Present

- Integrated **YOLOv11 (PyTorch)** pipeline in Python and C++ for real-time **autonomous object detection** on embedded systems travelling at 50+ km/h, achieving **98% detection accuracy** and **+30% mAP**
- Engineered a **metric learning pipeline** for target re-identification, computing mean vectors from Vision Transformer (**ViT**) embeddings to enable cosine similarity matching invariant to lighting and orientation
- Implemented **TensorRT FP16 quantization** to compress model weights, reducing memory bandwidth usage by **50%** while maintaining inference precision within **0.5%** of the baseline

Lead Developer | UBC Finds

Oct. 2025 – Present

- Architected **event-driven state synchronization** over Supabase realtime channels on a **React/Next.js** campus lost-and-found platform, eliminating race conditions through **atomic writes**

PROJECTS

Wealth Council | Python, FastAPI, React, Google Gemini, Fetch.ai

Mar. 2026

- Engineered an agentic investment platform, winning an award at the ProduHacks x Fetch.ai hackathon
- Architected a Multi-Agent Orchestration Layer using **Fetch.ai uAgents** to asynchronously compute portfolio risk metrics and integrated **FinBERT (NLP)** to perform sentiment analysis on live financial news streams
- Integrated Google Gemini to synthesize final reports, streamed to a React frontend in real time via **SSE**

GuardCam: AI Dashcam | Python, React Native, WebSockets, OpenCV

Mar. 2026

- Developed an edge-to-cloud driver monitoring system to detect micro-sleeps, utilizing a custom **TCP WebSocket bridge** to bypass HTTP overhead for low-latency video streaming
- Implemented **MediaPipe** and **OpenCV** on a headless Modal GPU server to map a 468-point 3D face-mesh, calculating Eye Aspect Ratio (EAR) to trigger sub-second haptic alerts

FPGA-Based Neural Network Inference Accelerator | Verilog, FPGA, C++, ARM

July 2025

- Designed and implemented a **single-layer neural network (MAC unit)** on a **DE1-SoC FPGA** using **Verilog**
- Developed a **memory-mapped C++ driver** on the ARM HPS for low-latency data streaming, offloading compute to the FPGA and enabling efficient hardware acceleration under **embedded Linux**
- Achieved **3,000 inferences/sec** with a **300%** performance speedup over ARM-only execution by optimizing **HPS-FPGA** communication

TECHNICAL SKILLS

Hardware/RTL: FPGA Design (Intel Quartus, QuestaSim, Platform Designer), Verilog, SystemVerilog, ARM SoCs

Languages: Python, C, C++, C#, SQL, Java, JavaScript, RISC-V & ARM Assembly

Technologies: Linux, PyTorch, TensorFlow, TensorRT, OpenCV, YOLO, React, Next.js, Supabase, PostgreSQL, Git